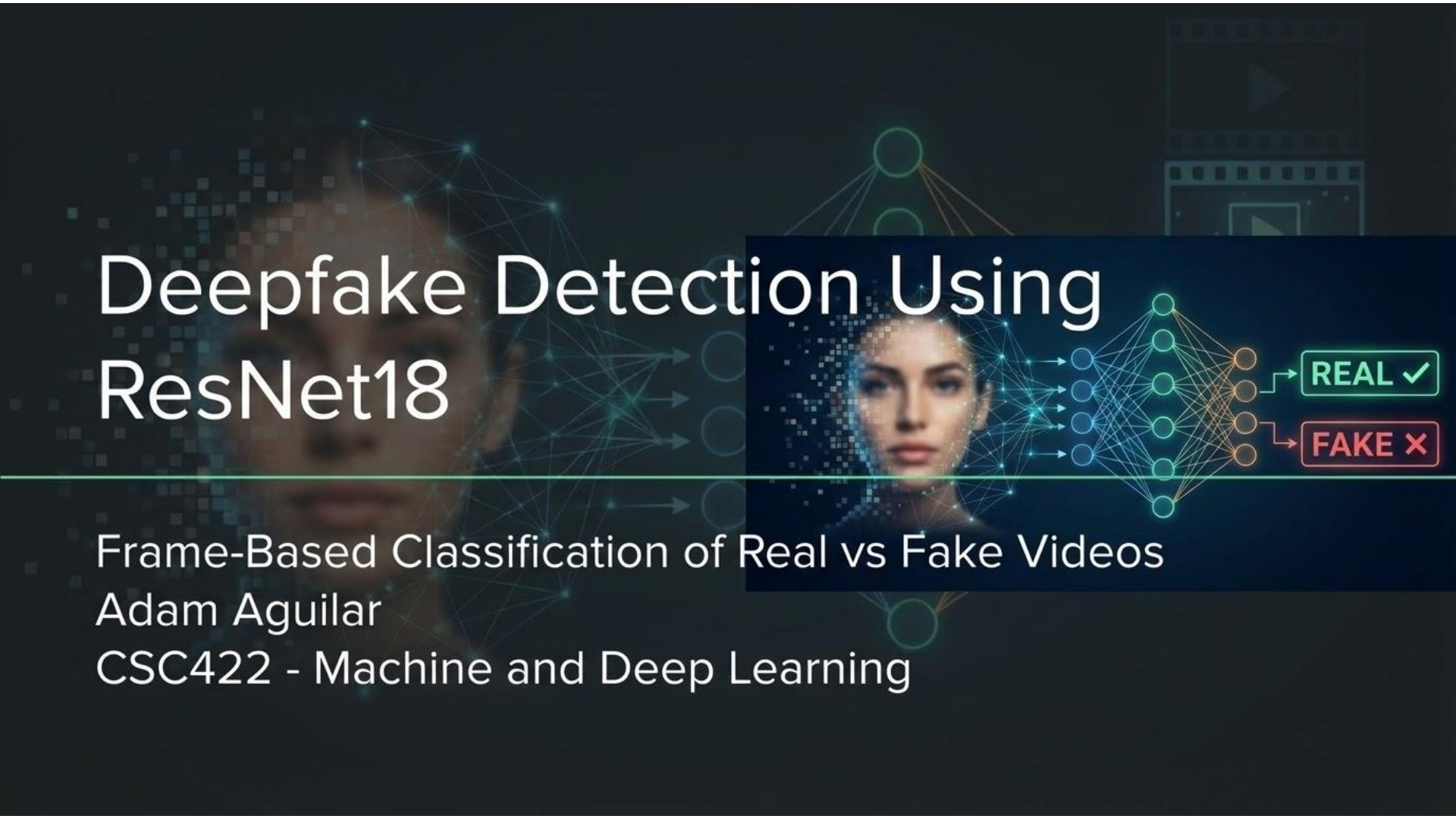


Deepfake Detection Using ResNet18



Frame-Based Classification of Real vs Fake Videos

Adam Aguilar

CSC422 - Machine and Deep Learning

Motivation

- Deepfakes are increasingly realistic
- Can spread misinformation
- Need reliable detection methods
- Goal: Classify video as REAL or FAKE



Dataset Overview

- Kaggle FaceForensics++ Deepfake Detection
- 400 labeled videos (REAL/FAKE)
- Extracted 6,800 frames for training
- Balanced: Fake $\approx 5,491$ | Real = 1,309

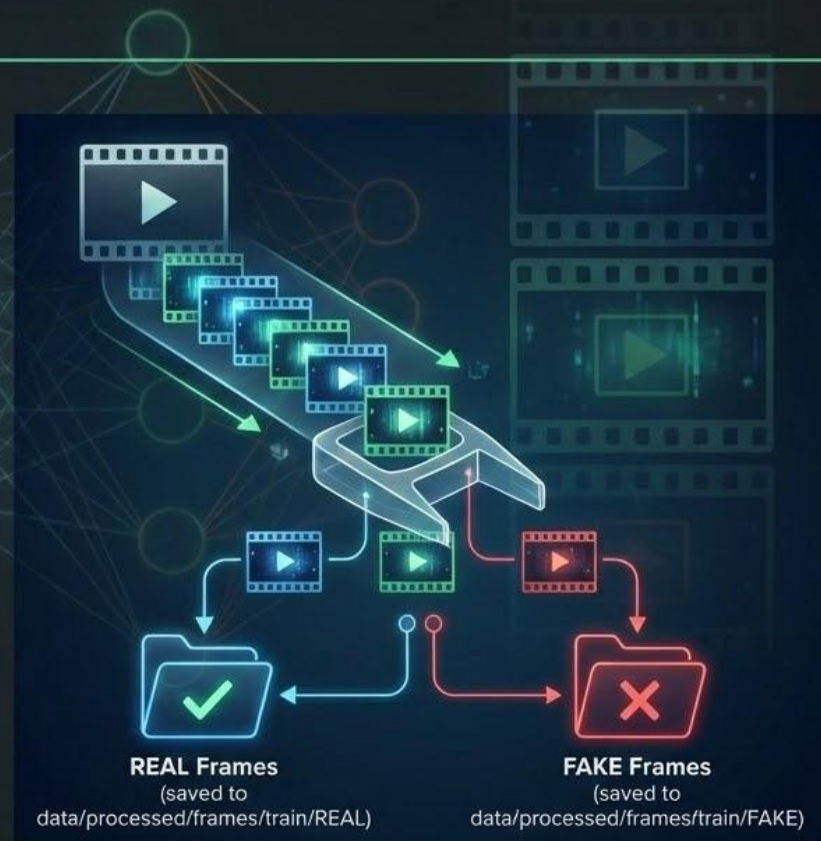


Project Pipeline



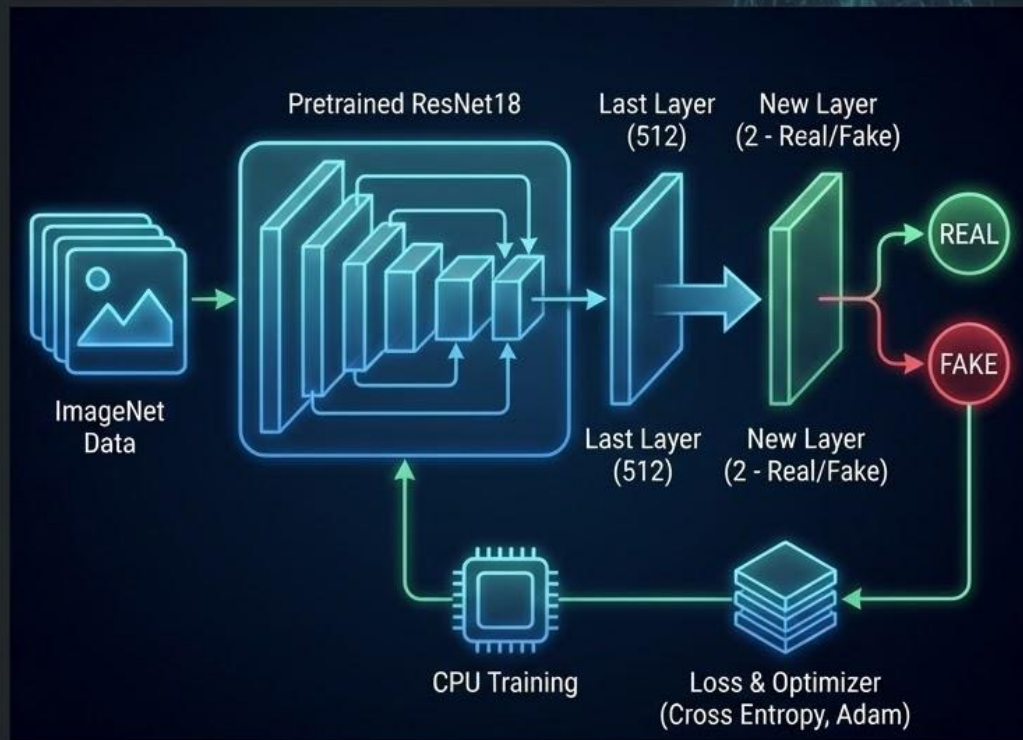
Frame Extraction

- Extract 16 frames per video
- Uniformly spaced
- Saved to data/processed/frames/train/REAL or FAKE
- Total frames: 6800 (including original single frame extraction)



Model Architecture (ResNet18)

-  Pretrained on ImageNet
-  Replaced last layer: 512 $\rightarrow$ 2
-  Trained on CPU
-  Loss: Cross Entropy
-  Optimizer: Adam



Training Setup & Validation Accuracy

- 🔗 Train/Validation split: 80/20
- 📁 Batch size: 16
- 🕒 Epochs: 5
- 🔄 Random augmentations
- 📈 Tracking accuracy + loss curves
- ✅ Final Validation Accuracy: 82.6%

```
Epoch 5/10
  Train loss: 0.1866  acc: 0.922
  Val   loss: 0.2282  acc: 0.900

Epoch 6/10
  Train loss: 0.1584  acc: 0.938
  Val   loss: 0.2544  acc: 0.904

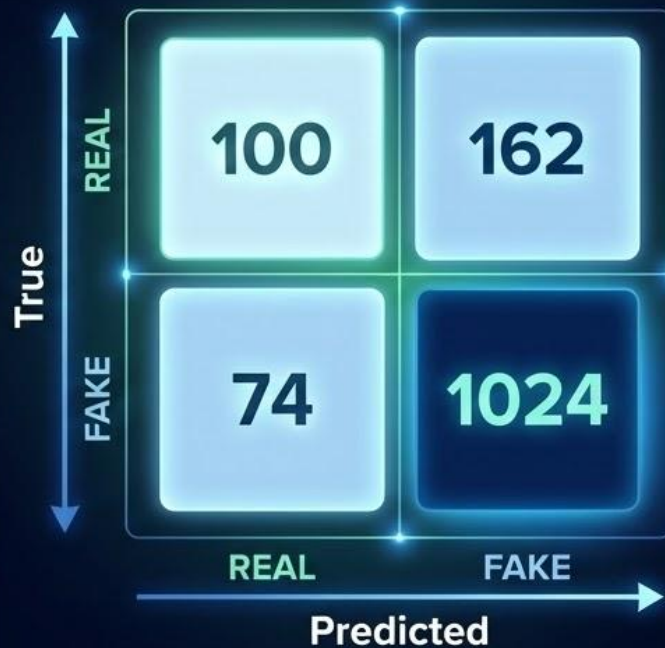
Epoch 7/10
  Train loss: 0.1388  acc: 0.947
  Val   loss: 0.1874  acc: 0.930

Epoch 8/10
  Train loss: 0.1224  acc: 0.957
  Val   loss: 0.1976  acc: 0.921

Epoch 9/10
  Train loss: 0.1161  acc: 0.958
  Val   loss: 0.1747  acc: 0.929

Epoch 10/10
  Train loss: 0.1099  acc: 0.958
  Val   loss: 0.1974  acc: 0.921
```

Confusion Matrix (Validation Set)



Classification Report



Fake detection is **very strong**:

Precision: 0.86



Recall: 0.93



F1 score: 0.89



Real detection is **weaker**:

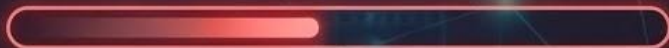
Precision: 0.57



Recall: 0.38



F1 score: 0.46

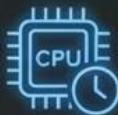


Limitations & Future Work

⚠ Limitations



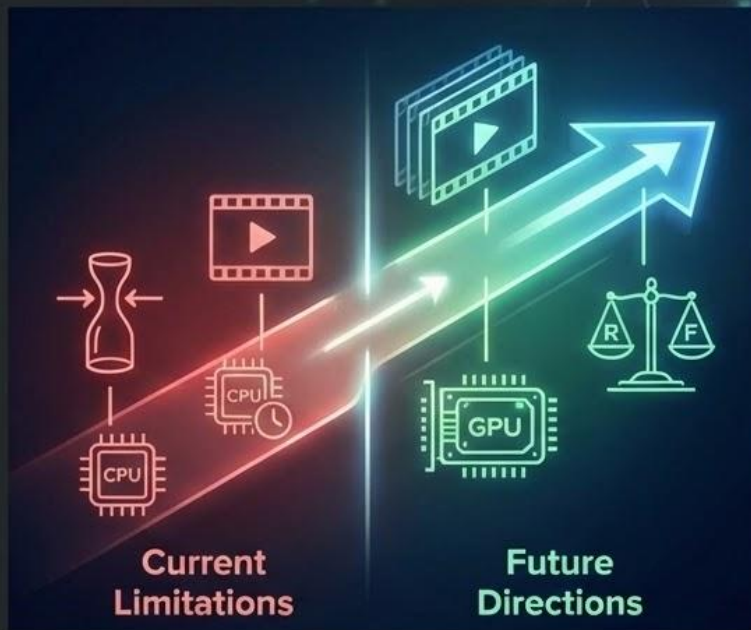
- Harder to detect REAL vs FAKE balance



- CPU training slows experimentation



- Only frame-level detection (not video)



📈 Future Work



- Try video-based models (3D CNN, LSTM)



- Improve balance between real/fake sample $\leq$ sample



- Add more augmentations